# The shadow that changed

Shadows at Noon

Name _____ Date _____



Little Wonderer found a long shape beside a standing stick. It was a shadow: a place where the stick blocked sunlight, the light from the sun. “Will it stay this long?” Little asked. Big set a linen notebook beside the garden path. They had a question to keep.



Little took a piece of chalk, a soft stick for making marks. “First mark the tip,” said Big. Little drew a small line where the shadow ended. To measure is to find how long something is. They used a measuring stick from the wooden stick’s base to the chalk.



At noon, the middle of the day, they came back. The wooden stick had stayed in place, but its shadow had changed. “It is short and crisp!” said Little. Around noon, the sun is near its highest point for the day. The morning shadow had reached much farther.

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|---|---|
| **noon** | the middle of the day, when the sun is near its highest point |
| **shadow** | a place where an object blocks light |
| **measure** | find a length using a unit, such as a small step or a centimeter |
| **chalk** | a soft stick that leaves a mark on a path or board |
| **stick** | a thin piece of wood |
| **sunlight** | light that comes from the sun |

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“Which way does it point?” asked Big. Little studied the ground. Sunlight reached one side of the stick. The shadow stretched away on the other side. The bright side faced the sun; the shadow pointed away from the light. Little traced its edge along the ground with one finger.



Little marked the new tip with chalk. They used the same measuring stick to check both lengths. They began at the same standing stick and used the same unit each time. The noon length was shorter. “The stick stayed the same,” Little said. “The sun’s height changed.”



In the notebook, Little drew a long line for morning and a short line for noon. Later, when the sun was lower, the shadow would grow longer again. A short noon shadow was a clue to a high sun. “What will we notice tomorrow?” Little asked.

1. Why was the same stick’s shadow shorter at noon?
2. Which side of the stick has the shadow, compared with the light?
3. How could you compare a morning shadow with a noon shadow?

## Your turn

Write one command sentence about measuring a shadow.

Example: Measure from the stick’s base to the shadow tip with the same unit.